Literal certificates for twenty-four atlas structure labels

L054 bounded recovery component

Source and scope. Connor and Leemans, *An atlas of subgroup lattices of finite almost simple groups*, Ars Math. Contemp. 8 (2015), 259–266, Section 3, distinguish an unspecified extension $N.Q$, a nonsplit extension $N \cdot Q$, a semidirect product $N : Q$ and a direct product $N \times Q$. Unparenthesized products associate to the left. The authors explicitly permit nonunique descriptions; their Figure 1 also uses an identification database and choices of desirable normal subgroups. We verify the displayed existence assertions, without claiming to reproduce those historical choices or to give canonical names. This is an elementary certificate method, not a novelty claim.

The inputs are the retained *current downloaded* companion PDFs for A_5 and M_{11}, both internally dated 25 November 2012, and their exact row maps to previously saved literal subgroup representatives. Their identity with every file available at the fixed 2013 edition is not assumed. All three table pages and both pages of *How to read the atlas* were directly inspected. In particular, $2^2, 3^2$ are native superscripts; they were not inferred from flattened strings "22" or "32". The finite roster below is the entire claim of this component.

Proposition 1 (Elementary verification rules)**.** *Let H be a finite group whose complete elements and multiplication are given literally. The following data certify the indicated assertions.*

1. *A generating element of order m certifies C_m. For $p = 2, 3$, commuting elements of order p whose exponent vectors give a bijection $\{0, \dots, p-1\}^a \to H$ certify C_p^a.*
2. *A bijective multiplication-preserving map from the standard finite permutation model certifies A_4, A_5 or S_3. Generators r, s of a group of order $2m$, with $|r| = m$, $|s| = 2$ and $srs = r^{-1}$, certify D_{2m}. Generators i, j of a group of order eight, both of order four, satisfying $i^2 = j^2$ and $j^{-1}ij = i^{-1}$ certify Q_8.*
3. *Actual subgroups $N, K \leq H$ with $N \lhd H$, $N \cap K = 1$ and $\langle N, K \rangle = H$ certify $H = N : K$. If they commute, they certify $H = N \times K$. Their two factor labels can be certified recursively.*
4. *If $N \lhd H$ has index two, its quotient is C_2. The extension is nonsplit if and only if every element of $H \setminus N$ has order different from two.*

Proof. The first assertions give explicit cyclic or elementary abelian coordinate isomorphisms. In the second, multiplication preservation and bijectivity give an isomorphism. The dihedral relations reduce every word to r^a or $r^a s$, so there are at most $2m$ elements; generation and the stated order give the dihedral presentation exactly. The quaternion relations similarly reduce words to i^a or $i^a j$ for $0 \leq a < 4$, and give the usual order-eight quaternion presentation. Normality in the third assertion gives $NK = H$, and trivial intersection makes $(n, k) \mapsto nk$ bijective. Multiplication is the semidirect multiplication for the actual conjugation action; when the factors commute it is direct multiplication. Finally, a section of $H \to H/N \cong \mathrm{C}_2$ exists exactly when the nonidentity quotient element has an outside lift of order two. This requires checking all outside elements: a single badly chosen section proves nothing. □

Proposition 2 (Reuse by explicit transport)**.** *A literal bijective homomorphism from one certified concrete finite model to an actual subgroup transports its certified structure. A shared registry may therefore store one such model and its certificate, with a separately checked isomorphism for each occurrence. A registry key need not, and must not, assert that the displayed label determines a unique isomorphism type.*

Proof. An isomorphism preserves products, orders, normality, intersections and generation. It transports complements and the entire outside-element list. Thus every hypothesis of the preceding proposition is preserved, inductively through the label tree. In the implementation each registry record has a concrete literal origin and generators. Every transport is a full bijection checked on every multiplication pair. Distinct concrete models with the same tree may have separate registry records. □

Theorem 1 (The bounded atlas roster)**.** *Under the acquired literal row maps, all nine A_5 labels and the following fifteen M_{11} labels are true of their actual saved subgroup representatives:*

row	*structure*	*row*	*structure*
12	$11:5$	13	$Q_8:S_3$
14	$(3:S_3)\cdot 2$	16	$S_3\times S_3$
17	$2^2:S_3$	18	$Q_8:3$
19	$D_{10}\cdot 2$	20	$3:S_3$
21	$S_3\times 3$	22	$Q_8:2$
27	3^2	28	Q_8
29	8	30	D_8
36	2^2		

The A_5 rows, in order, are $A_5, A_4, D_{10}, S_3, 5, 2^2, 3, 2, 1$. Here a bare integer denotes a cyclic group, and the dihedral subscript denotes its order.

Proof. The certificate files give complete actual permutation elements and original subgroup generators for each of the twenty-four rows. An independent token parser reads all thirty-nine retained M_{11} subgroup constructors and joins the chosen fifteen exactly to those generators. For A_5, the lexicographically sorted list of all sixty even permutations gives the indexing of the acquired literal subgroup array; all nine element lists are joined exactly. The accepted numerical row maps, the native label transcriptions and these actual-generator joins are checked separately. No source label is used to change a row map or to filter an ambient census.

The verifier closes each displayed generating list and compares it to the complete listed elements. It computes products by literal permutation composition. Each certificate is then checked by the preceding rules, with its exact native label tree. This pays forty-four factor nodes, 4,032 standard-model multiplication pairs and all twenty-eight outside elements of the two index-two nonsplit assertions. The outside lists are exhaustive, and the actual two quotient cosets are retained. Each colon node also records its actual nontrivial-action Boolean; each direct node checks commutation. The shared registry has twenty-four concrete models for forty-four factor occurrences, giving twenty reuses; all 15,671 transport multiplication pairs are explicitly checked. These finite certificates supply every hypothesis of the two propositions. □

Finite evidence and limits. The bounded witness discovery used only these already retained small subgroups (order at most sixty), finished in less than one second, and performed no ambient subgroup census. Its exact inputs, code and terminal receipt are preserved. Discovery is not a verification premise: the final checker uses only the literal certificates, original subgroup arrays, source joins and multiplication rules. Eighteen damaged examples are rejected, including the nonnormal diagonal S_3 in $S_3\times S_3$, noncommuting proposed direct factors, a nonhomomorphic bijection, and an index-two subgroup in $\mathrm{C}_4\times\mathrm{C}_2$ with a bad chosen section but an outside involution. The same checks pass with Python optimization enabled.

The numerical class catalogue and its completeness retain their separately accepted interfaces; this proof adds the stated labels only. It does not certify any other row, the atoms M_9, M_{10}, larger classical aliases, additive-exponent conventions, or the historical naming software. There is no whole-atlas, whole-original, conjecture, Lean or measurement credit. The current author supplied this proof and implementation; root proposed the shared concrete-model registry and authorized the narrow pilot. Distinct review is required before this component is acquired.